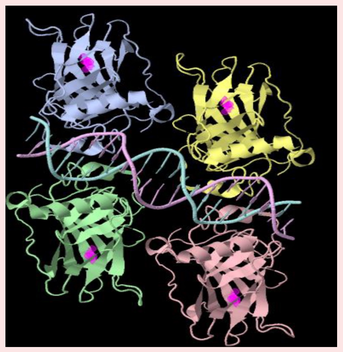
P53-V157F: Too Cool to Transactivate? A Chill, Temperature-Sensitive Mutant on the Fritz!

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Stanford University, BIO 45, Sec03_3A, Tuesday

Introduction

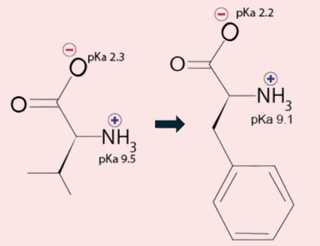
p53: One of the Most Mutated Genes in Cancer

p53 plays a pivotal role as a tumor suppressor gene by preventing the proliferation of potentially cancerous cells. p53 functions mainly by halting cell division to allow for DNA repair or initiating apoptosis if the damage is irreparable. When DNA damage is detected, p53 can activate the expression of a variety of genes, including the cyclin-dependent kinase inhibitor p21 and the pro-apoptotic protein Bax.[1,2]



V157F Mutation in p53

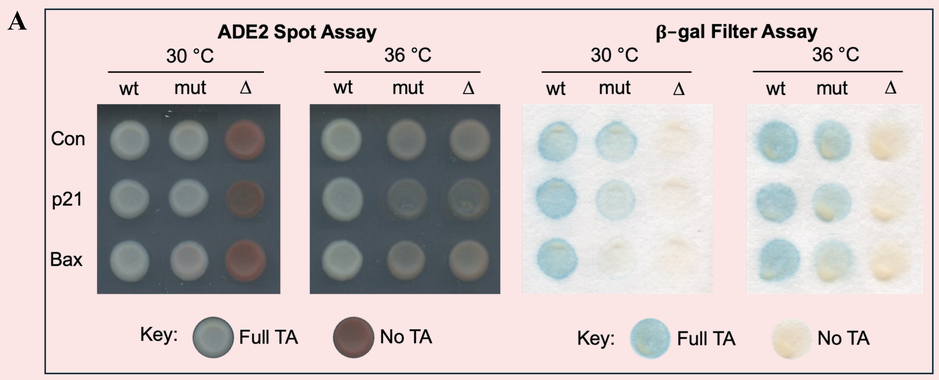
The V157F mutation has been identified as a hotspot in lung squamous cell carcinomas and is linked to both loss of tumor suppressor capability and a gain-of-function that promotes cancer progression and resistance to therapy.[3] The V157F mutation occurs in the DNA-binding domain of p53. Molecular dynamics simulations suggest that this alteration results in a structure with slightly lesser β-sheet structure and increased tendencies toward aggregation compared to wildtype p53, which may be due to increases in the protein's volume and exposed surface area.[3]



V157F Mutation

Results

Temperature-Sensitive Transactivation Defect



B

p53-RE	Temp (°C)	Overall Transactivation Score	
		Ours	Group
Con	30	3	2.3
	36	1.5	1.5
p21	30	2.5	2
	36	1.5	1.3
Bax	30	2	1.5
	36	1.5	1.2

C

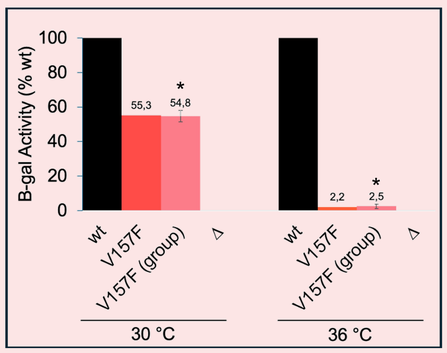


Figure 1. p53-V157F has a temperature-sensitive transactivation defect. A. Qualitative assays show a slight reduction in transactivation of p53-RE-driven ADE2 and lacZ reporter genes by p53-V157F compared to a wildtype p53 at 30 °C, with a more pronounced decrease at 36 °C, suggesting a temperature-sensitive impairment. **B.** Quantification of combined qualitative assay results from (A), with Transactivation Scores assigned as follows: 3 = equivalent to wildtype (full transactivation), 1 = equivalent to deletion (no transactivation), and 2 = intermediate reduction in transactivation. **C.** Kinetic β-galactosidase assay reveals that p53-V157F maintains moderate transactivation of a Consensus p53-RE-driven lacZ gene compared to a wildtype p53 at 30 °C but shows a marked loss of activity at 36 °C, confirming a temperature-sensitive impairment in p53-V157F function.

Temperature-Sensitive Stability Reduction

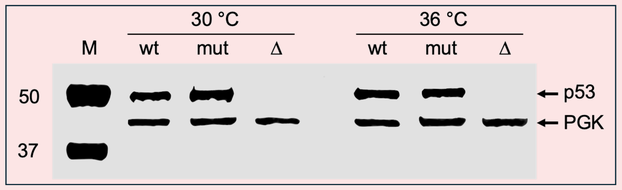


Figure 2. p53-V157F has temperature-sensitive decrease in stability. Western Blot shows yeast extract expressing p53-V157F has comparable protein level (121%) compared to its control of wildtype p53 that were grown at 30 °C and statistically significant reduced protein level (66%) at 36 °C, suggesting a temperature-sensitive decrease in stability, through SDS-PAGE, electrotransfer, and immunodetection where PGK served as a loading control. This trend is also observed in the group data, confirming temperature sensitivity (96% vs. 73%).

Cold-Sensitive DNA-Binding Defect

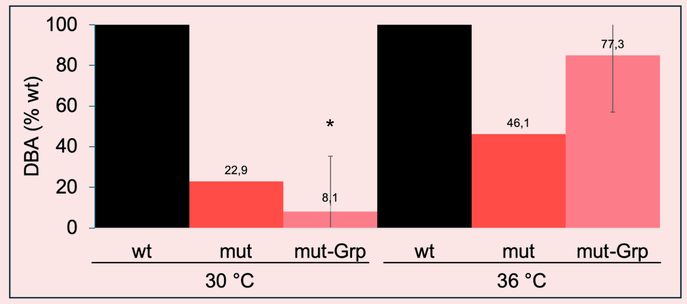
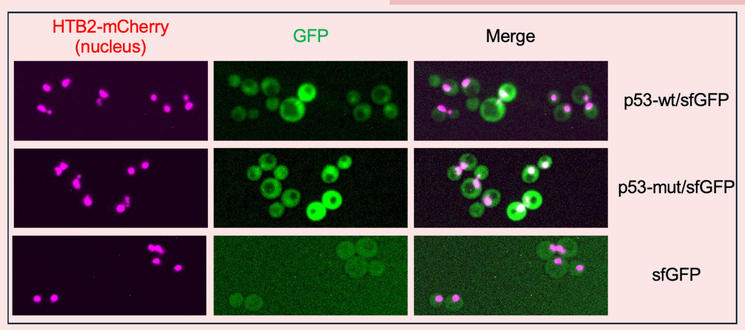


Figure 3. p53-V157F has a cold-sensitive DNA-binding defect. DNA binding assay shows yeast extract expressing p53-V157F has low DNA binding efficiency compared to its control of wildtype p53 at 30 °C and moderately reduced (partial restoration) efficiency at 36 °C, suggesting a cold-sensitive decrease in binding efficiency. It was quantified colorimetrically by measuring the development of color that correlates with the binding of an HRP-linked secondary antibody that eventually binds to the p53 proteins attached to biotinylated Consensus p53-RE DNA fragment.

Nuclear Localization Defect



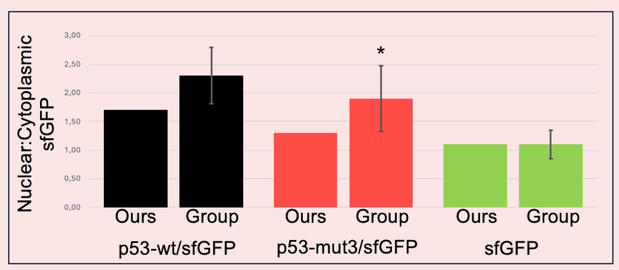


Figure 4. p53-V157F has a nuclear localization defect. Fluorescent microscopy shows recombinant yeast cells expressing p53-V157F/sfGFP has slightly reduced nuclear localization (80%) compared to wildtype p53 grown at 30 °C. Nuclear:Cytoplasmic sfGFP ratio was quantified to assess localization efficiency, HTB2-mCherry serving as a nuclear marker.

Conclusions

- Qualitative and quantitative assays shows p53-V157F has a transactivation defect across p53-REs (Con, p21, Bax) at both 30 °C and 36 °C.
- Western Blot indicates p53-V157F protein levels are comparable to wildtype at 30 °C but drop significantly at 36 °C.
- DNA-Binding Assay shows at 30 °C, p53-V157F exhibits low DNA binding efficiency relative to wildtype, with partial restoration of binding observed at 36 °C.
- Fluorescent microscopy reveals that p53-V157F is less efficiently localized to the nucleus compared to wildtype p53 at 30 °C.
- **p53-V157F is a multifaceted mutant with defects in its stability, nuclear localization, and DNA binding, likely contributing to its loss of tumor suppressor function.**

Future Directions

- Structural Studies via Cryo-EM or X-ray Crystallography to determine the precise structural changes caused by the V157F mutation.
- FRET-Based RNA Polymerase II Recruitment Assay by designing p53-CFP and RNA Pol II-YFP fusion proteins to measure whether p53-V157F impairs the recruitment of RNA Pol II to p53 target promoters in live cells.

References

1. Rocca V, Blandino G, D'Antona L, Iuliano R, Di Agostino S. Li–Fraumeni Syndrome: Mutation of TP53 Is a Biomarker of Hereditary Predisposition to Tumor: New Insights and Advances in the Treatment. Cancers. 2022; 14(15):3664.
2. Vousden, K., Lu, X. Live or let die: the cell's response to p53. Nat Rev Cancer 2, 594–604 (2002).
3. Lei J, Li X, Cai M, Guo T, Lin D, Deng X, Li Y. Insights into Allosteric Mechanisms of the Lung-Enriched p53 Mutants V157F and R158L. International Journal of Molecular Sciences. 2022; 23(17):10100.